

TOPIC FIVE

ENZYMES

LECTURE 7

Learning Objectives

- 1) Define Enzymes in chemical terms
- 2) Classify Enzymes into two major groups according to the Nature of Enzymes
- 3) List the importance of Enzymes in biochemical processes .
- 4) Explain Properties of Enzyme and Regulation of enzyme activity
- 5) Explain mechanism of action of enzymes(Hypothesis of action of enzymes on substrate)
- 6) Explain biological importance/functions of Enzymes
- 7) Explain the various Factors Affecting Enzyme Activity
- 8) Describe the two major types of enzyme inhibition and their roles in enzyme activity
- 9) Describe diseases/disorder associated with Enzymes

Introduction

- Enzymes are protein catalysts for chemical reaction in biological systems. They increase the rate of chemical reactions taking place within living cells without changing themselves.
- Enzymes are usually proteins (some RNA)
 - In general, names end with suffix “ase”
 - Enzymes are catalysts
 - increase the rate of a reaction
 - not consumed by the reaction
 - act repeatedly to increase the rate of reactions
 - Enzymes are often very “specific” – promote only 1 particular reaction
 - Reactants also called “substrates” of enzyme

Why Enzymes?

- Accelerate and control the rates of vitally important biochemical reactions
- Greater reaction specificity
- Milder reaction conditions
- Capacity for regulation
- Enzymes are the agents of metabolic function.
- Metabolites have many potential pathways
- Enzymes make the desired one most favorable

Nature of Enzymes

- Most enzymes are protein in nature. Depending on the presence and absence of a nonprotein component with the enzyme can exist as, simple enzyme or holoenzyme
- **1. Simple enzyme:** It is made up of only protein molecules not bound to any nonproteins.
 - Example: Pancreatic Ribonuclease.
- **2. Holo enzyme** is made up of protein groups and non-protein component.
 - The protein component of this holo enzymes is called **apoenzyme**
 - The non-protein component of the holo enzyme is called a **cofactor**. I

Nature of Enzymescoenzyme

- If this cofactor is an organic compound it is called a **coenzyme** and if it is an inorganic groups it is called **activator**. (Fe $2+$, Mn $2+$, or Zn $2+$ ions).
- If the cofactor is bound so tightly to the apoenzyme and is difficult to remove without damaging the enzyme it is sometimes called a **prosthetic group**

Coenzymes

- **Coenzymes** are derivatives of vitamins without which the enzyme cannot exhibit any reaction. One molecule of coenzyme is able to convert a large number of substrate molecules with the help of enzyme.
 - Coenzyme accepts a particular group removed from the substrate or donates a particular group to the substrate
 - Coenzymes are called co substrate because the changes that take place in substrates are complimentary to the changes in coenzymes.
 - The coenzyme may participate in forming an intermediate enzyme-substrate complex
- Example: **NAD, FAD, Coenzyme A**
- **Metal ions in enzymes**
- Many enzymes require metal ions like **Ca²⁺, K⁺, Mg²⁺, Fe²⁺, Cu²⁺, Zn²⁺, Mn²⁺ and Co²⁺** for their activity.
- Metal-activated enzymes-form only loose and easily dissociable complexes with the metal and can easily release the metal without denaturation.

Properties of Enzyme

1. Active site

- Enzyme molecules contain a special pocket or cleft called the active site. The active site contains amino acid chains that create a three-dimensional surface complementary to the substrate.

2. Catalytic efficiency/ Enzyme turnover number

- Typically each enzyme molecule is capable of transforming 100 to 1000 substrate molecule in to product each second.
- Enzyme turn over number refers to the amount of substrate converted per unit time (carbonic anhydrase is the fastest enzyme).

Properties of Enzyme

3. Specificity

- Enzymes are specific for their substrate. Specificity of enzymes are divided into:
- a. Absolute specificity:- this means one enzyme catalyzes or acts on only one substrate. For example: Urease catalyzes hydrolysis of urea but not thiourea.
- b. Stereo specificity- some enzymes are specific to only one isomer even if the compound is one type of molecule:

Properties of Enzyme

4. Regulation

- Enzyme activity can be regulated- that is, enzyme can be, activated or inhibited so that the rate of product formation responds to the needs of the cell.

Classification of Enzymes

Enzymes are classified on the basis of the reactions they catalyze- Each enzyme is assigned a four-digit classification number and a systematic name, which identifies the reaction catalyzed.

- **Class I. Oxidoreducases-** Enzymes catalyzing oxidation reduction reactions.
- **Class II. Transferases:** Enzymes catalyzing a transfer of a group other than hydrogen (methyl, acyl, amino or phosphate groups)

Classification of Enzymes

- **Class III. Hydrolases:** Enzymes catalyzing hydrolysis of ester, ether, peptido, glycosyl, acid-anhydride, C-C, C-halide, or P-N-bonds by utilizing water.
- **Class IV. Lyases:** Enzymes that catalyze removal of groups from substances by mechanisms other than hydrolysis, leaving double bonds.
- **Class V. Isomerases:**
 - Includes all enzymes catalyzing interconversion of optical, geometric, or positional isomers.
- **Class VI. Ligases or synthetases.**
 - Enzymes catalyzing the linking together of 2 compounds coupled to the breaking of a pyrophosphate bond in ATP or similar trinucleotides: GTP, UTP etc. included are enzymes catalyzing reactions forming C-O, C-S, C-N, and C-C bonds,

MECHANISM OF ACTION OF ENZYMES

1. Lock and Key Model:

- **Emil Fischer's model Lock and Key model 1890**
developed a model called Lock: Key model of enzyme action implies that the active site of the enzyme is complementary in shape to that of its substrate, i.e. the shape of the enzyme molecule and the substrate molecule should fit each other like a lock and Key

2. Induced Fit Model (Hand in Glove analogy)

- **In 1958, Daniel Koshland**, postulated another model; which implies that the shapes & the active sites of enzymes are complementary to that of the substrate only after the substrate is bound.

MECHANISM OF ACTION OF ENZYMES

Mechanism of Enzyme Action (1913)

- Michaelis and Menten have proposed a hypothesis for enzyme action, which is most acceptable. According to their hypothesis, the enzyme molecule (E) first combines with a substrate molecule (S) to form an enzyme substrate (ES) complex which further dissociates to form product (P) and enzyme (E) back. Enzyme once dissociated from the complex is free to combine with another molecule of substrate and form product in a similar way.

Factors Affecting Enzyme Activity

1. **Temperature**; Starting from low temperature as the temperature increases to certain degree the activity of the enzyme increases because the temperature increase the total energy of the chemical system .
2. **PH**; Extreme pH can also lead to denaturation of the enzyme, because the structure of the catalytically active protein molecule depends on the ionic character of the amino acid chains.

Factors Affecting Enzyme Activity

3. Substrate/enzyme concentration etc.

An increase in the substrate concentration increases the enzyme activity till a maximum is reached. Further increase in substrate concentration does not increase rate of reaction because the substrate molecule combine with all available enzyme molecules at their active site till not more active sites are available (The active Sites become saturated). At this state the enzyme is obtained **it maximum rate (V max)**.

Enzyme Inhibition

- Any substance that can diminish the velocity of an enzyme-catalyzed reaction is called an inhibitor and the process is known as inhibition.
- There are two major types of enzyme inhibition, **Irreversible** and **Reversible**.

Irreversible Inhibition

- The type of inhibition that can not be reversed by increasing substrate concentration or removing the remaining free inhibitor.
 - Eg. Diisopropyl & fluorophosphate (DFP) Inhibits the enzyme acetyl cholinesterase, important in the transmission of nerve impulses.
- Acetyl cholinesterase catalyzes the hydrolysis of Acetylcholin (to acetic acid and choline) a neurotransmitter substance functioning in certain portions of the nervous system

Reversible Inhibition

- This type of inhibition can be:
 - **Competitive,**
 - **Non-competitive and**
 - **uncompetitive**

Competitive Inhibition: This type of inhibition occurs when the inhibitor binds reversibly to the same site that the substrate would normally occupy, therefore, competes with the substrate for that site.

Non-Competitive Inhibition

- In non-competitive inhibition the inhibitor binds at different site rather than the substrate-binding site.

Reversible Inhibition cont....

When the inhibitor binds at this site there will be a change in conformation of the enzyme molecules, which leads to the reversible inactivation of the catalytic site.

Uncompetitive Inhibition

- Uncompetitive Inhibitor binds only to ES complex at locations other than the catalytic site. Substrate binding modifies enzyme structure, making inhibitor-binding site available. Inhibition cannot be reversed by substrate. In this case apparent V_{max} and K_m decreased.

Regulation of enzyme activity

1. Irreversible covalent Activation / Zymogen activation

- Some enzymes are secreted in an inactive form called Proenzymes or zymogens.
- At the site of action specific peptide bonds are hydrolysed either enzymatically or by PH changes to convert it into active form, e.g. Pepsinogen to pepsin, Trypsinogen to trypsin, plasminogen to plasmin.
- After hydrolysis when it is activated, it cannot be reconverted into proenzyme form.

Regulation of enzyme activity

2. Reversible Covalent Modification

- By addition of or removal of phosphate or adenylate, certain enzymes are reversibly activated and inactivated as per the requirement.
- Protein kinase of muscle phosphorylate phosphorylase kinase, glycogen synthetase by making use of ATP.

Regulation of enzyme activity

3. Allosteric Modulation

- In addition to simple enzymes that interact only with substrates and inhibitors, there is a class of enzymes that bind small, physiologically important molecules and modulate activity in ways other than those described above. These are known as **allosteric enzymes**; the small regulatory molecules to which they bind are known as **effectors**.
- Allosteric effectors bring about catalytic modification by binding to the enzyme at distinct allosteric sites, well removed from the catalytic site, and causing conformational changes that are transmitted through the bulk of the protein to the catalytically active site(s).
- The hallmark of effectors is that when they bind to enzymes, they alter the catalytic properties of an enzyme's active site.
- Those that increase catalytic activity are known as positive effectors. Effectors that reduce or inhibit catalytic activity are negative effectors.

Regulation of enzyme activity

4. Feedback inhibition

- In allosteric regulation in which end products inhibit the activity of the enzyme is called "feedback inhibition". The kinetics of feedback inhibition can be competitive, mixed, etc. It is the commonest way of regulation of a biosynthetic pathway. Feedback regulation generally occurs at the earliest functionally irreversible step unique in the biosynthetic pathway.

CLINICAL SIGNIFICANCE OF ENZYMES

Measurement of enzymes concentration of mostly the latter type in plasma gives valuable information about disease involving tissues of their origin.

- ❖ The plasma lipase level may be low in liver disease, Vitamin A deficiency, some malignancies, and diabetes mellitus. It may be elevated in acute pancreatitis and pancreatic carcinoma.
- ❖ The main use of amylase estimations is in the diagnosis of acute pancreatitis. The plasma amylase level may be low in liver disease and increased in high intestinal obstruction, mumps, acute pancreatitis and diabetes.

CLINICAL SIGNIFICANCE OF ENZYMES

- ❖ Trypsin is secreted by pancreas. Elevated levels of trypsin in plasma occur during acute pancreatic disease.
- ❖ The alkaline phosphates are a group of enzymes, which hydrolyze phosphate esters at an alkaline pH. In bone the enzyme is found in osteoblasts and is probably important for normal bone function. The level of these enzymes may be increased in rickets and osteomalacia, hyperparathyroidism, paget's disease of bone, obstructive jaundice, and metastatic carcinoma.
I,,,,,,,,,,,,,mSerum alkaline phosphatase levels may be increase in congestive heart failure result of injury to the liver.

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MANY THANKS